Raw Materials, Compositions, Equipment, and Manufacturing Processes for Lip Seal Manufacture

1. Introduction

Lip seals, also known as rotary shaft seals or oil seals, are critical components that prevent lubricant leakage and exclude contaminants in rotating machinery such as water pumps, compressors, and automotive engines. The manufacture of lip seals involves the precise selection of elastomeric materials, carefully formulated rubber compounds, specialized molding equipment, and multi-stage production processes that transform raw rubber into a precision sealing element with a thin, flexible sealing lip.

Modern lip seals are predominantly manufactured from hydrogenated nitrile rubber (HNBR), EPDM, or fluororubber, selected based on the operating environment and the fluid to be sealed . The sealing lip itself may incorporate specialized fillers to control surface roughness and improve hydrodynamic lubrication .

2. Raw Materials and Compositions

2.1 Elastomeric Base Materials

The base elastomer determines the fundamental chemical and thermal resistance of the lip seal. The primary elastomeric materials used in lip seal manufacture include:

Hydrogenated Nitrile Rubber (HNBR): HNBR is widely used for lip seals in water pumps and compressor applications due to its excellent oil resistance, heat resistance, and mechanical strength . HNBR grades with varying acrylonitrile (AN) content are often blended to optimize properties. For example, a blend of medium-to-high-nitrile HNBR (AN content 30–38%) and high-nitrile HNBR (AN content 40–46%) in a 35:65 to 50:50 weight ratio provides an optimal balance of oil resistance and low-temperature flexibility .

EPDM (Ethylene Propylene Diene Monomer): EPDM is used for water pump seals and applications requiring excellent weather and water resistance .

Fluororubber (FKM): Fluororubber is selected for high-temperature applications and resistance to aggressive fluids .

Nitrile Rubber (NBR): Standard NBR remains in use for general-purpose oil seals .

2.2 Reinforcing Fillers

Reinforcing fillers provide mechanical strength, hardness, and tear resistance to the rubber compound.

Carbon Black: The most common reinforcing filler, typically used at 30–70 parts by weight per 100 parts of rubber . Carbon black grades such as G-SO are commonly employed in HNBR lip seal compounds .

Silica: Precipitated silica with specific BET surface area (40–100 m^2/g) is used in compressor lip seal compounds where low compression set and oil resistance are critical . Silica is typically used at 40–80 parts by weight .

2.3 Non-Reinforcing Fillers

Non-reinforcing fillers serve multiple functions in lip seal compounds: they control surface roughness of the sliding surface, cut deposits formed by lubricant additives, and form a liquid film to improve lubrication .

Aluminum Silicate: With an average particle diameter of approximately 5.3 μm, aluminum silicate is used at 5–90 parts by weight, preferably 5–70 parts . The presence of these coarse

particles creates surface roughness (Ra 1–30 μm) on the sealing lip, which improves hydrodynamic lubrication and prevents deposit accumulation .

Calcium Silicate: Used as an alternative to aluminum silicate, with particle diameters of approximately 4.5 μm .

Other Non-Reinforcing Fillers: Magnesium silicate, carbon fiber, iron oxide, titanium oxide, and diatomaceous earth may also be used, with average particle diameters of 1–40 μm .

2.4 Coupling Agents

Coupling agents strengthen the adhesion between the rubber matrix and the filler, suppressing the collection of LLC (long-life coolant) at the rubber/filler interface due to osmosis .

Silane-Based Coupling Agents: Preferred for lip seal compounds, including vinyl-, glycidoxy-, methacryloxy-, and amino-based silanes . Specific examples include 3-glycidoxypropyltrimethoxysilane and N-2-(aminoethyl)-3-aminopropylmethyldimethoxysilane. The typical proportion is 0.1–5 parts by weight, preferably 0.5–3 parts .

2.5 Crosslinking Systems

Organic Peroxides: Used as crosslinking agents for HNBR, EPDM, and fluororubber. Examples include dicumyl peroxide, α,α'-di(t-butylperoxy)diisopropylbenzene, and 2,5-dimethyl-2,5-di(t-butylperoxy)hexane . The proportion is typically 0.5–10 parts by weight, preferably 1–8 parts .

Co-Crosslinking Agents: Polyfunctional unsaturated compounds such as triallyl isocyanurate (TAIC), trimethylolpropane trimethacrylate (TMPT), or ethylene glycol dimethacrylate are used at 1–15 parts by weight, preferably 2–10 parts . Co-crosslinking agents promote tight crosslinking and suppress softening and swelling caused by coolant osmosis .

2.6 Processing Aids and Other Additives

Acid Acceptors: Zinc oxide (typically 5 parts) and magnesium oxide are used to neutralize acidic byproducts and act as curing components .

Antioxidants: N-1,3-dimethylbutyl-N'-phenyl-p-phenylenediamine is commonly used at approximately 3 parts by weight to protect against thermal and oxidative degradation .

Processing Aids: Stearic acid, palmitic acid, and paraffin wax may be added to improve processing .

Plasticizers: Ester-based plasticizers may be added at up to 10 parts by weight for compressor lip seal applications .

2.7 Representative Formulation

A representative HNBR compound for water pump lip seals (Example 1 from EP2703650A1) :

Component	Material	Parts by Weight
Base elastomer	HNBR (Zetpol 2011)	100
Reinforcing filler	Carbon black (G-SO)	45
Non-reinforcing filler	Aluminum silicate (5.3 μm)	15
Coupling agent	Silane-based (KBM602)	0.5
Co-crosslinking agent	Ethylene glycol dimethacrylate	6
Antioxidant	N-1,3-dimethylbutyl-N'-phenyl-p-phenylenediamine	3

| Organic peroxide | α,α'-di(t-butylperoxy)diisopropylbenzene | 3 |

3. Equipment

3.1 Mixing Equipment

Internal Mixers: Intermix, kneader, or Banbury mixers are used for initial mastication and mixing of the rubber compound .

Open Mills (Two-Roll Mills): A 10-inch open roll is commonly used for final compounding and sheeting. The roll gap is adjusted to achieve the desired sheet thickness (typically 0.1–2 mm) .

3.2 Preforming Equipment

Preforming Dies: Matched metal dies are used to prepare preformed rings of the compounded rubber. For spring-loaded seals, the garter spring may be pre-vulcanized with a thin rubber layer before final molding to stabilize its position .

3.3 Molding Equipment

Compression Molding Presses: Hydraulic presses with heated platens are used for vulcanization molding. Mold temperatures are typically 150–200°C .

Injection Molding Machines: Rotary injection machines with multiple stations (e.g., 14-station "Speedy Mould") are used for high-volume production, suitable for nitrile, acrylic, viton, silicone, and fluorinated rubbers .

Transfer Molding Equipment: For more complex geometries, transfer molding may be employed.

3.4 Finishing Equipment

Cryogenic Machining Lathes: For precision trimming of the sealing lip, the molded seal stock is cooled to render the elastomer rigid and machined with controlled cutting tools. Coolants such as refrigerated Oleum may be used to achieve temperatures as low as −80°F .

Grinding Equipment: Plunge grinding may be used to achieve final surface finish on the sealing lip, eliminating lead marks from machining .

Deflashing Equipment: Mechanical or cryogenic deflashing removes flash from molded parts.

3.5 Quality Control Equipment

Electromechanical Inspection Systems: Automated equipment checks inner diameter (ID), outer diameter (OD), eccentricity, lip pressure, and presence of garter springs. Lip pressure is measured using a spring-loaded split mandrel, and spring presence is verified by magnetic sensors .

4. Manufacturing Processes

4.1 Compounding and Mixing

The raw materials are weighed according to the formulation and mixed in an internal mixer or on an open mill. A typical mixing sequence involves:

1. Mastication of the base elastomer

2. Addition of reinforcing fillers and coupling agents

3. Addition of non-reinforcing fillers

4. Addition of crosslinking agents, co-agents, and processing aids

The compound is sheeted to the desired thickness on an open mill . For HNBR compounds, the mixed compound is typically conditioned for 24 hours before molding.

4.2 Preforming

The compounded rubber is cut into preforms of appropriate weight and shape for the mold cavity. For spring-loaded lip seals, the garter spring is pre-treated: cleaned with gasoline, dried, coated with a release agent, and wrapped with a thin layer of rubber compound (0.1–0.2 mm thickness) . The wrapped spring is then pre-vulcanized in a matched die at reduced pressure (approximately 60% of full cure pressure) and reduced time (approximately 30% of full cure time) to stabilize the spring position .

4.3 Molding and Vulcanization

The preform (and pre-vulcanized spring, if applicable) is placed in the mold cavity along with the metal case (if the seal is metal-cased). The mold is closed and heated under pressure to vulcanize the rubber.

Typical Vulcanization Conditions:

- **Temperature**: 150–200°C

- **Time**: 3–60 minutes

- **Pressure**: Sufficient to ensure complete mold filling

A representative two-stage cure for HNBR lip seals:

- **Primary vulcanization**: 180°C for 5 minutes

- **Secondary (oven) vulcanization**: 150°C for 1 hour

During molding, the rubber flows to fill the cavity, forming the sealing lip geometry. The mold may incorporate ridges or protrusions to form hydrodynamic channels on the lip surface . Excess rubber may flow into overflow grooves and is subsequently trimmed .

4.4 Post-Curing (Secondary Vulcanization)

After demolding, the seals undergo secondary vulcanization in an oven to complete the crosslinking reaction and optimize physical properties. Typical conditions are 150°C for 1 hour .

4.5 Finishing Operations

Deflashing: Flash formed at mold parting lines is removed by mechanical trimming or cryogenic deflashing.

Lip Machining (Optional): For precision applications, the sealing lip may be machined to exact dimensions. The molded seal is cooled to render the elastomer rigid and machined on a lathe with controlled cutting parameters. For Buna rubber, machining is conducted at temperatures between −30°F and −40°F with a cutting tool rake angle of 20–30° and clearance angle of 15–25° . Plunge grinding may follow to achieve the final surface finish.

Surface Roughness Control: The presence of non-reinforcing filler (average particle diameter ≥1 μm) creates surface roughness (Ra 1–30 μm) on the sliding surface, which is critical for hydrodynamic lubrication and deposit cutting .

4.6 Quality Control and Inspection

Finished lip seals undergo automated inspection including:

- Inner and outer diameter measurement

- Eccentricity check

- Lip pressure verification

- Garter spring presence confirmation

Additional testing may include:

- **Physical property evaluation**: Elongation at break ≥150%

- **Torque test**: Measurement of rotational torque with water as sealing fluid

- **Immersion test**: Volume change after 2000 hours in LLC solution at 120°C; volume change <+10%

- **Deposition test**: Evaluation of deposit formation on shaft after 50 hours at 6000 rpm, 120°C, 0.15 MPa

- **Sealing performance**: Leakage rate <0.2 ml/hour

5. Summary Table

| Stage | Materials | Equipment | Conditions |

|-------|-----------|-----------|------------|

| Compounding | HNBR/EPDM/FKM, carbon black, silica, aluminum silicate, silane coupling agent, peroxide, co-agent | Internal mixer, open mill | Room temperature, roll gap 1–2 mm |

| Preforming | Compounded rubber, garter spring | Preforming die | Spring pre-vulcanized at 60% pressure, 30% time |

| Molding | Preform + case/spring | Compression press or injection machine | 150–200°C, 3–60 min |

| Secondary Cure | Molded seal | Oven | 150°C, 1 hour |

| Finishing | Cured seal | Lathe, grinder | Cryogenic machining at −30 to −40°F |

| Inspection | Finished seal | Electromechanical system | ID/OD, eccentricity, lip pressure, spring presence |

6. Summary

The manufacture of lip seals is a precision process requiring careful selection of elastomeric materials and fillers, specialized mixing and molding equipment, and rigorous quality control. HNBR remains the material of choice for water pump and compressor lip seals due to its excellent oil and heat resistance. The incorporation of non-reinforcing fillers with controlled particle size is critical for achieving the surface roughness necessary for hydrodynamic lubrication and deposit cutting. The two-stage vulcanization process (primary molding and secondary oven cure) ensures complete crosslinking and optimal physical properties. Automated inspection systems verify dimensional accuracy and sealing performance before the seals are released for use.